MIL-S-13257H(AR)
12 February 1980
SUPERSEDING
MIL-S-13257G (MU)
10 January 1972

MILITARY SPECIFICATION

SIGNALS, ILLUMINATION, GROUND RED STAR, M126A1,WHITE STAR, M127A1 AND GREEN STAR, M195, PARACHUTE, PARTS AND LOADING, ASSEMBLING AND PACKING

This specification is approved for use by the US Army Armament Research and Development Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the parts and the loading, assembling and packing of three signals of different colors, designated as Signal, Red Star Parachute: M126A1: Signal, White Star Parachute: M127A1: and Signal, Green Star Parachute: M195.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-223	-	Powder, Black
MIL-C-10464	-	Cans, Hermetic Sealing, Metal, Light Gage, Tear-Strip Type
MIL-W-12332	-	Welding, Resistance, Spot and Projection, for Fabricating Assemblies of Low Carbon Steel
MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

FEDERAL

FED-STD-151	-	Metals, Test Methods

FSC: 1370

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY

MIL-STD-1167 - Ammunition Data Cards.
MIL-STD-1234 - Pyrotechnics; Sampling, Inspection and Testing

DRAWINGS (See 6.5)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND (ARRADCOM)

PRODUCT AND PACKAGING DRAWINGS

7548414	-	Container, Ammunition, Metal M492 for Ground Signals
7548415	-	Box, Packing, Ammunition for Ground Signals, in Metal Container, M492
8797968	-	Signals, Illumination, Ground Red Star, M126A1 and White Star M127A1 Parachute Signal Assembly
9255112	-	Signal, Illumination, Ground Green Star, Parachute, M195 Assembly

INSPECTION EQUIPMENT DRAWINGS

9201136	-	Tunnel, Light
9201268	-	Procedure, Light Output Measurement
9201390	-	Procedure, Photocell Checkout
9201392	-	Procedure, Color Value Measurement
9247071	-	Photocell

(Copies of specifications, standards and drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

3. REQUIREMENTS

3.1 Materials. Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Assembly. The assembly shall comply with all requirements specified on Drawing (Dwg.) 8797968 and 9255112 and with all requirements specified in applicable specifications.

3.3 Functioning.

3.3.1 Premature burst. The assembly shall not burst within one hundred (100) feet of the launcher. The illuminant assembly shall not eject at an altitude of less than one hundred (100) feet from the ground within a horizontal distance of two hundred and fifty (250) feet of the launcher, or at an altitude of less than fifty (50) feet from the ground at a horizontal distance in excess of two hundred and fifty (250) feet from the launcher.

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3.3.2 Parachute. The parachute shall not separate from the assembly, delay opening from time of ejection for more than 5 seconds, or remain partially open.

3.3.3 Hot temperature (68 to 74°C or 155 to 165°F). The signal assembly shall function and comply with the requirement of 3.3.1 when tested as specified in 4.5.2.1.

3.3.4 Ambient temperature (18 to 24°C or 65 to 75°F). The signal assemblies shall function and comply with the following requirements when tested as specified in 4.5.2.2:

a. The requirements of 3.3.1 and 3.3.2
b. The illuminant assembly shall burn for not less than 50.0 seconds in the M126A1 and the M195 assemblies, and for not less than 25.0 seconds in the M127A1 assembly.
c. The illuminant assembly shall not eject at an angle of more than 30 degrees from the vertical, or emit the first trace of light at an altitude of less than 500 feet.
d. The average altitude for all illuminant assemblies which function above 500 feet shall be not less than 725 feet.
e. The average angle from the vertical for all illuminant assemblies which function within the maximum (max) angle from the vertical (30 degrees) shall be not more than 12 degrees.

3.3.5 Cold temperature (-57 to -51°C or -70 to -60°F). The signal assemblies shall function and comply with the following requirements when tested as specified in 4.5.2.3:

a. The requirements of 3.3.1 and 3.3.2
b. The illuminant assembly burning time shall comply with 3.3.4b.
c. The average angle from the vertical of the ejected illuminant assemblies shall be not more than 25 degrees. The altitude shall be recorded for informational purposes.

3.4 Leakage. The sealed metal container shall show no evidence of leakage when tested as specified in 4.5.3.

3.5 Container tear strip. The container shall comply with the requirements of MIL-C-10464 when tested as specified in 4.5.4.

3.6 First article inspection. This specification makes provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.7 Cracks. The tube casing shall be free of cracks under visual inspection after hydrostatic testing.

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3.8 Workmanship. All parts shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, rust and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection.

b. Quality Conformance Inspection.

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of items in sample quantities as indicated in Table 1.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table 1 specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I - FIRST ARTICLE INPSECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-13257H(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
	Signals, Illumination, Ground Star Parachute				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Inert parts (See appropriate TDPL for drawings)				
	Examination for defects	Note 1	—	3.2	Visual/Gage
	Propellant grain (Dwg. 8887529)				
	Overall density	80	—	3.1	4.5.5.1
	Sectional density	Note 2	—	3.1	4.5.5.2
	Containers (Dwg. 7548414)				
	Leakage Test	96	—	3.4	4.5.3
	Tear Strip Test (from the Leakage Test)	96	—	3.5	4.5.4
	Illuminant Assemblies (9255113, 9295010, 9328576)				
	Static burn tests	20 ea	—	3.3	4.5.10
	Signals, Assemblies (8797968 or 9255112)				
	Functioning, hot	32	—	3.3.3	4.5.2.1
	Functioning, ambient	32	—	3.3.4	4.5.2.2
	Functioning, cold, fixture fired	16	—	3.3.5	4.5.2.3.1
	Functioning, cold, hand fired	16	—	3.3.5	4.5.2.3.2

NOTES:
1. Twenty-five (25) complete sets of each component and inert subassembly shall be submitted for inspection and tests in accordance with the specification and drawing requirements.
2. 40 grains shall be selected from the total 80 sample for the sectional density test.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, each inspection lot of signal assemblies shall contain:

a. Parts of one kind from one supplier
b. Primers of one interfix lot number
c. Black powder from not more than one lot
d. Signal assemblies from not more than one designation
e. Composition ingredients from not more than one lot
f. Illuminant composition produced by one manufacturer under one continuous set of operating conditions and which consists of one or more batches that have been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-13257H(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.1	Tube Casing				8797921
					NEXT HIGHER ASSEMBLY
					8797968 or 9255112
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Hydrostatic pressure test		100%	3.2	4.5.6
2	Hydrostatic letter "H" indicator missing		100%	3.2	Visual
3	Presence of cracks		100%	3.7	Visual
Major					
101	Concentricity of ends including perpendicularity		0.40%	3.2	Gage
102	Total length, max		0.40%	3.2	Gage
103	Inside diameter of ends, max		0.40%	3.2	Gage
104	Inside diameter of body, max		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-13257H(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Bolt				8797928
					NEXT HIGHER ASSEMBLY
					8797968 or 9255112
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length, min		0.40%	3.2	Gage
102	Pitch diameter of thread, min		0.40%	3.2	Gage
103	Major diameter of thread, min		0.40%	3.2	Gage
104	Diameter of flash holes, min		0.40%	3.2	Gage
105	Length under head		0.40%	3.2	Gage
106	Diameter of head, min		0.40%	3.2	Gage
107	Concentricity of head with thread		0.40%	3.2	Gage
108	Flash holes obstructed or fail to intersect each other		0.40%	3.2	Visual
109	Bolt pull strength test	(a)		3.2	4.5.7
Minor					
201	Length from head to intersecting flash hole, min		0.65%	3.2	Gage
202	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:
(a) 200 samples/lot, acc 1, rej 2.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-13257H(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER 8797929
4.4.2.3	Barrel, Rocket		NEXT HIGHER ASSEMBLY 8797968 or 9255122

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Crack		100%	3.2	Visual
2	Draw marks, inclusions, scratches more than .015 deep in sidewall		100%	3.2	Visual-Gage
Major					
101	Diameter of primer hole, max		0.40%	3.2	Gage
102	Concentricity of primer hole and counterbore with the inside diameter		0.40%	3.2	Gage
103	Inside diameter		0.40%	3.2	Gage
104	Depth of primer counterbore		0.40%	3.2	Gage
105	Outside diameter of rear end		0.40%	3.2	Gage
106	Length to knurl, max		0.40%	3.2	Gage
Minor					
201	Wall thickness, min		0.65%	3.2	Gage
202	Thickness through bottom of charge cavity		0.65%	3.2	Gage
203	Diameter of charge cavity		0.65%	3.2	Gage
104	Total length		0.65%	3.2	Gage
205	Length to top of charge cavity		0.65%	3.2	Visual
206	Radii missing		0.65%	3.2	Visual
207	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS MIL-S-13257H(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8797947
4.4.2.4	Tail Assembly				NEXT HIGHER ASSEMBLY 8797968 or 9255112
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length, max		0.40%	3.2	Gage
102	Small width, max		0.40%	3.2	Gage
103	Inside diameter of upper tail ring, min		0.40%	3.2	Gage
104	Perpendicularity of ends with diameter		0.40%	3.2	Gage
105	Large width, max		0.40%	3.2	Gage
106	Assembly damaged or distorted to the extent that function will be impaired		0.40%	3.2	Visual
107	Weld missing		0.40%	3.2	Visual
108	Weld flash excessive		0.40%	3.2	Visual
109	Perpendicularity of fins with each other		0.40%	3.2	Gage
110	Parallelism of opposite fins		0.40%	3.2	Gage
111	Weld peel strength test	(a)		3.2	4.5.11
112	Hardness test (See Dwg. 8797951)	(b)		3.2	4.5.8
Minor					
201	Length from ring to tail vane, min		0.65%	3.2	Gage
202	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:
(a) 32 samples/lot, acc 0, rej 1.
(b) Twenty (20) tail vanes shall be selected from each heat treated batch for hardness testing.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-13257H(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8797953
4.4.2.5	Firing Cap & Spring Clip Assembly				NEXT HIGHER ASSEMBLY 8797968 or 9255112
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter over rivet head, max		0.40%	3.2	Gage
102	Rivet head protrudes inside wall		0.40%	3.2	Visual
103	Firing pin missing or loose		0.40%	3.2	Visual - Manual
104	Spring clip missing or looae		0.40%	3.2	Visual - Manual
105	Push Test - Firing cap and spring assembly (prior to insertion of primer)	(a)		3.2	4.5.9
Minor					
201	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:
(a) 200 samples/lot, acc 1, rej 2.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-13257H(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8797954
4.4.2.6	Pin, Firing				NEXT HIGHER ASSEMBLY 8797953
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Profile of point		0.40%	3.2	Gage
102	Length from point to shoulder		0.40%	3.2	Gage
Minor					
201	Protective coating damaged exposing base metal		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.00%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-13257H(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.7	Cap, Firing		8797955
			NEXT HIGHER ASSEMBLY 8797953

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Concentricity of cap with firing pin recess		0.40%	3.2	Gage
102	Depth of firing pin recess		0.40%	3.2	Gage
Minor					
201	Length of spring clip recess, min		0.65%	3.2	Gage
202	Diameter of firing pin recess		0.65%	3.2	Gage
203	Diameter of rivet hole		0.65%	3.2	Gage
204	Evidence of poor workmanship		1.00%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-13257H(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.8	Signal Body & Delay Assembly		8797961
			NEXT HIGHER ASSEMBLY: 8797968 or 9255112

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
101	Marking misleading or unidentifiable		0.65%	3.2	Visual
102	Designation incorrect		0.65%	3.2	Visual
103	Evidence of poor workmanship		1.00%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-13257H(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.9	Parachute Assembly				8797991
					NEXT HIGHER ASSEMBLY 8797968 or 9255112
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Parachute torn, contains holes or is improperly tied to suspension cord or improperly attached to chain bead assembly, when applicable		0.40%	3.2	Visual
102	Shrouds twisted		0.40%	3.2	Visual
Minor 201	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.10	Grain Propellant				8887529
					NEXT HIGHER ASSEMBLY 8887530
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Crack (4 power magnification)		100%	3.2	Visual
2	Overall Density Test		100%	3.1	4.5.5.1
3	Sectional Density Test	(b)		3.2	4.5.5.2
4	Propellant Ballistic Test	(c)		3.3	4.4.3.3
Major					
101	Inside diameter		0.40%	3.2	Gage
102	Outside diameter		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES: a. Grains shall be assembled and loaded by same manufacturing methods as intended for production.
b. 25 samples to be tested; accept 0, reject 1.
c. 150 grains to be assembled and tested in 50 signals; accept 0, reject 1.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-13257H(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8887530
4.4.2.11	Propellant Assembly				NEXT HIGHER ASSEMBLY 8797968 or 9255112
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Pettman cement coating on interior surface where not permitted		100%	3.2	Visual
2	Pettman cement coating insufficient (slight imperfections of pin hole type or those resulting from the bristles of the brush used in applying the cement, are acceptable)		100%	3.2	Visual
3	Grain broken		100%	3.2	Visual
Major					
101	Length, max		0.40%	3.2	Gage
102	Glaze not removed from interior surfaces		0.40%	3.2	Visual
Minor					
201	Sheath damaged		0.65%	3.2	Visual
202	Marking misleading or unidentifiable		0.65%	3.2	Visual
203	Evidence of poor workmanship		1.00%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-13257H(AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.12	Plate, Exhaust	1	1	9235026
				NEXT HIGHER ASSEMBLY: 8797968 or 9255112

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Exhaust hole missing		100%	3.2	Gage
Major					
101	Diameter at large end of taper		0.40%	3.2	Gage
102	Diameter at small end of taper		0.40%	3.2	Gage
103	Thickness, min		0.40%	3.2	Gage
104	Length of flange, max		0.40%	3.2	Gage
105	Diameter of flange		0.40%	3.2	Gage
106	True position of bolt hole with diameter under flange including perpendicularity		0.40%	3.2	Gage
107	True position of exhaust holes with diameter under flange		0.40%	3.2	Gage
108	Diameter under flange, min		0.40%	3.2	Gage
109	Diameter of bolt hole, min		0.40%	3.2	Gage
Minor					
201	Thickness, max		0.65%	3.2	Gage
202	Diameter of bolt hole counterbore, max		0.65%	3.2	Gage
203	Depth of bolt hole counterbore, max		0.65%	3.2	Gage
204	Finish improper		0.65%	3.2	Visual
205	Evidence of poor workmanship		1.00%	3.8	Visual

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CLASSIFICATION OF DEFECTS & TESTS MIL-S-13257H(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9255113, 9295010 or 9328576
4.4.2.13	Illuminant Assembly				NEXT HIGHER ASSEMBLY 8797968 or 9255112
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Color designation incorrect		100%	3.2	Visual
Major					
101	Outside diameter, max		0.40%	3.2	Gage
102	Quickmatch missing or not partially embedded in the priming charge, when applicable		0.40%	3.2	Visual/Manual
103	Candle power and color value test (a)	(b)	--	3.2	Test/4.5.10
Minor					
201	Assembly damaged		0.65%	3.2	Visual
202	Marking misleading or unidentifiable		0.65%	3.2	Visual
203	Priming charge on exposed portion of quickmatch (where applicable)		0.65%	3.2	Visual
204	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES: (a) This test is for acceptance of illuminant assembly lots prior to use in production.
(b) 20 samples/lot or batch - same sample for both tests. Candle power: acc 1, rej 2; Color value: acc 0, rej 1.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-13257H(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.14	Delay Assembly				9251412
					NEXT HIGHER ASSEMBLY 8797961
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	Delay charge missing		100%	3.2	Visual
Major 101	Delay composition above flush or more than max. below flush		0.40%	3.2	Gage
102	Flash hole obstructed		0.40%	3.2	Visual
Minor 201	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-13257H(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9251411
4.4.2.15	Housing, Delay				NEXT HIGHER ASSEMBLY 8797961
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter under flange		0.40%	3.2	Gage
102	Small outside diameter		0.40%	3.2	Gage
103	Pitch diameter of thread, max		0.40%	3.2	Gage
104	Minor diameter of thread, max		0.40%	3.2	Gage
105	Diameter of delay cavity		0.40%	3.2	Gage
106	Depth of delay cavity		0.40%	3.2	Gage
Minor					
201	Diameter of flange		0.65%	3.2	Gage
202	Length of diameter under flange		0.65%	3.2	Gage
203	Location of flash hole		0.65%	3.2	Gage
204	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.16	Assembly (signal body and delay assembly after assembly of tail assembly, propellant assembly exhaust plate and bolt)	1 OF 1	8797968 or 9255112
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Tail assembly binds on tube casing		100%	3.2	Visual (See Note)
2	Exhaust plate flange missing or inadequate to engage tail assembly		100%	3.2	Visual (See Note)
Major					
101	Bolt or tube casing loose		0.40%	3.2	Manual
102	Bolt flash hole obstructed		0.40%	3.2	Visual
103	Propellant assembly loose		0.40%	3.2	Manual
Minor					
201	Evidence of poor workmanship		1.00%	3.8	Visual
	NOTE: Starting with the upper tail ring contacting the signal body and delay assembly, if the tail assembly does not slide down the end of the tube casing by its own weight when the signal is held, forward end up, in a vertical position, the assembly shall be classed defective and removed from the lot. If tail assembly falls off because of absence or inadequacy of the exhaust plate flange, the assembly shall also be classed defective and removed from the lot.				

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-H-13257H(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
.4.2.17	Assembly (signal body and delay assembly after assembly of tail assembly, propellant assembly, exhaust plate and bolt), but prior to inserting illuminant assembly and washer disc assembly)	1 OF 1	8797968 or 9255112
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Weight of expelling charge, min (See Note)		0.40%	3.2	Balance
102	Expelling charge missing		0.40%	3.2	Visual
Minor 201	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES: Expelling charge shall be weighed just prior to insertion of charge in signal body and delay assembly.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-H-13257H(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.18	Assembly (signal body and delay assembly after assembly of tail assembly, propellant assembly, exhaust plate and bolt, washer disc assembly and illuminant assembly, but prior to inserting parachute in body)	1 OF 1	8797968 or 9255112 NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Suspension cord insecurely tied to illuminant assembly		0.40%	3.2	Visual
102	Component missing or damaged to extent that function may be impaired or chain bead assembly improperly fastened to illuminant assembly		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-H-13257H(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 8797968 or 9255112
4.4.2.19	Assembly, prior to assembling rocket assembly in rocket barrel				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	Weight of initiating charge, min (See note)		100%	3.2	Balance
Major 101	Rocket barrel washer missing		0.40%	3.2	Visual
102	Primer missing or not sealed		0.40%	3.2	Visual
103	Initiating charge missing		0.40%	3.2	Visual
Minor 201	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:
The initiating charge shall be weighed just prior to insertion of charge in rocket barrel.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-H-13257H(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.20	Assembly, after assembling rocket assembly but prior to inserting seal, rocket barrel				8797968 or 9255112
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Tape missing, insufficient, or incorrectly applied on cork seal		0.65%	3.2	Visual
202	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-13257H(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.21	Assembly, prior to assembling firing cap and spring clip assembly		8797968 or 9255112
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Color of coating on rocket barrel seal incorrect		100%	3.2	Visual
Major					
101	Rocket barrel seal missing		0.40%	3.2	Visual
102	Thickness of lacquer coating insufficient except that pin holes and surface roughness caused by the cork surface contour are acceptable		0.40%	3.2	Visual
103	Rocket barrel dented		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.22	Signal, Illum, Ground Parachute, Assembly				8797968 or 9255112
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Designation incorrect		100%	3.2	Visual
2	Firing cap assembled on primer end		100%	3.2	Visual
Major					
101	Total length, max		0.40%	3.2	Gage
102	Label missing		0.40%	3.2	Visual
103	Assembly damaged to the extent that function will be impaired		0.40%	3.2	Visual
Minor					
201	Marking misleading or unidentifiable		0.65%	3.2	Visual
202	Contents loose		0.65%	3.2	Manual
203	Evidence of poor workmanship		1.00%	3.8	Visual
204	Knurl not color coated		0.65%	3.2	Visual

NOTES:
Ballistic functioning test to be performed as specified in 4.4.3.2.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-S-13257H(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 7548414
4.4.2.23	Container				NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Prior to sealing				
Critical	None defined				
Major					
101	Signal inverted		0.40%	3.2	Visual
102	Packing component missing		0.40%	3.2	Visual
	Container, sealed				
Critical					
1	End color missing or incorrect		100%	3.2	Visual
2	Embossing missing or incorrect		100%	3.2	Visual
Major					
103	Key missing		0.40%	3.2	Visual
104	Container damaged		0.40%	3.2	Visual
105	Leakage test		100%	3.4	4.5.3
106	Tear strip test	(a)		3.5	4.5.4
Minor					
201	Marking misleading or unidentifiable		0.65%	3.2	Visual
202	Contents loose		0.65%	3.2	Manual
203	Protective coating damaged exposing base metal		0.65%	3.2	Visual
204	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES: (a) 80 samples/lot, acc 1, rej 2.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-S-13257H(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.24	Wood packing box				7548415
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Box damaged to extent that contents are exposed (See Note)		0.40%	5.2	Visual
102	Hardware or strapping missing, broken or loose		0.40%	5.2	Visual
103	Board broken		0.40%	5.2	Visual
Minor					
201	Contents loose		0.65%	5.2	Manual
202	Hardware or strapping improperly assembled		0.65%	5.2	Visual/Manual
203	Car seal missing unsealed or improperly engaged		0.65%	5.2	Visual/Manual
204	Marking misleading or unidentified		0.65%	5.2	Visual
205	Evidence of poor workmanship		1.00%	3.8	Visual

NOTES: This inspection is not intended to duplicate the original inspection of the boxes, but is to preclude acceptance of boxes which have been damaged in handling subsequent to manufacture.

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4.4.3 Production testing. Unless otherwise specified herein, the following tests shall be performed using MIL-STD-105 for sampling plan requirements.

4.4.3.1 Moisture content (See Table II) Major Defect.

TABLE II

Material	Drawing	Inspection Method Paragraph
Delay composition	9251412	4.5.1.4
Black Powder - for Green Star	9255112	4.5.1.1
- for Red Star or White Star	8797968	4.5.1.1
Illuminant composition - Green	9255113	4.5.1.3
- White	9295010	4.5.1.3
- Red	9328576	4.5.1.2
Cork	8797922	4.5.1.5
	8797923	4.5.1.5
Felt	8797937	4.5.1.5
	8797958	4.5.1.5
Paper	8797936	4.5.1.5
	8797938	4.5.1.5
	8797958	4.5.1.5
Chipboard	8797927	4.5.1.5
	8797980	4.5.1.5
Hardboard	8797938	4.5.1.5
Wood	8797941	4.5.1.5
Parachute assembly	8797991	4.5.1.5
Propellant assembly	8887530	4.5.1.6
Grain, propellant	8887529	4.5.1.6
Padding material (packing)	7548414	4.5.1.5

Note: The contractor shall provide adequate controls to insure that the materials comply with the requirement. For verification, the contractor shall select and subject to test, one sample of each material from each eight hours production. A composite sample shall not be used. If the moisture content of a sample exceeds the requirements and loading has not begun, the material represented by the sample shall be rejected. If components have been loaded with a material containing excessive moisture, the remaining unloaded material and the loaded components shall be rejected. Test procedures shall be as specified in 4.5.2.

4.4.3.2 Functioning. The signal assemblies shall be observed for any evidence of failure to comply with the requirements as classified in Table III when tested as specified in 4.5.2.

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TABLE III

Defect	Classifiaction
Premature burst (See 3.3.1)	Critical
Incorrect color (See Dwgs 9328576 & 9295010 & 9255113	Critical
Parachute failure (See 3.3.2)	Major
Burning time incorrect (See 3.3.4.b and 3.3.5b)	Major
Altitude less than min at +70°F temperature (See 3.3.4c)	Major
Angle from vertical greater than 30 degrees at +70°F temperature (See 3.3.4c)	Major
Average altitude under min at +70°F temperature (See 3.3.4d)	Major
Average angle from vertical over max at +70°F temperature (See 3.3.4e)	Major
Average angle from vertical over max at cold temperature (See 3.3.5c)	Major
Signal fails to function (See 3.3)	Major
Assembly bursts at distance of more than one hundred (100) feet from launcher (See 3.3.1)	Major

4.4.3.2.1 Hot temperature.

4.4.3.2.1.1 Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the acceptance criteria specified, one hundred (100) assemblies shall be selected for test. The lot shall be rejected if any signal assembly shows evidence of a critical defect during the test.

4.4.3.2.1.2 After three (3) consecutive lots have complied with the criteria of 4.4.3.2.1.1, twenty-five (25) assemblies shall be selected from each lot for test. The lot shall be rejected if any signal assembly shows evidence of a critical defect during the test.

4.4.3.2.2. Ambient temperature.

4.4.3.2.2.1 Beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the acceptance criteria specified, one hundred and twenty-five (125) assemblies shall be selected from each lot for test. The lot shall be rejected if during the test, a critical defect occurs, the average angle from the vertical exceeds the requirement, the average altitude is less than the requirement, or if eight (8) or more assemblies exhibit any of the remaining major defects listed in Table III.

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4.4.3.2.2.2 After three consecutive lots have complied with the criteria of 4.4.3.2.2.1, fifty (50) assemblies shall be selected from each lot for test. The lot shall be rejected if during the test, a critical defect occurs, the average altitude is less than required, the average angle from the vertical exceeds the requirement or if four (4) or more assemblies exhibit any of the remaining major defects listed in Table III (Para. 4.4.3.2).

4.4.3.2.3 Cold temperature, fixture fired.

4.4.3.2.3.1. Beginning with the first lot produced and continuing until three (3) consecutive lots have been found to be free of critical defects, thirty-two (32) assemblies shall be selected from each lot for test. The lot shall be rejected if during the test a critical defect occurs (See Table III).

4.4.3.2.3.2 After three (3) consecutive lots have complied with the criteria of 4.4.3.2.3.1, ten (10) assemblies shall be selected from each lot for this test. If any critical defect occurs the lot shall be rejected.

4.4.3.2.4 Cold temperature, hand fired.

4.4.3.2.4.1 Beginning with the first lot produced and continuing until three (3) consectuive lots have been found to be acceptable, one hundred and twenty-five (125) assemblies shall be selected from each lot for test. The lot shall be rejected if, during the test, a critical defect occurs, the average angle from the veritcal exceeds the requirement, or if eight (8) or more assemblies exhibit any of the remaining major defects listed in Table III. If the combined number of remaining major defects found in 4.4.3.2.2.1 and this paragraph equal or exceed twelve (12), the lot shall also be rejected. (See note after the following paragraph)

4.4.3.2.4.2 After three (3) consecutive lots have complied with the criteria of 4.4.3.2.4.1, fifty (50) assemblies shall be selected from each lot for test. The lot shall be rejected, if during the test, a critical defect occurs, the average angle from the vertical exceed the requirement or if 4 or more assemblies exhibit any of the remaining major defects listed in Table III. (See note)

NOTE: A star which ignites and burns normally but subsequently fails to meet minimum burning time requirement due to ground impact, will not be classified defective. Firing records will be annotated accordingly.

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4.4.3.3 Propellant ballistic test, Critical defect. 50 signals per propellant sub-lot of loose propellant composition are to be ballistically tested in fully loaded signals at hot temperature in accordance with 4.5.2.1 and 4.5.12. Upon successfully meeting the requirements as classified in Table III of 4.4.3.2 the propellant composition used and the method of manufacture of the propellant assembly shall be considered adequate for production. All production will then utilize the spacer, propellant part no. 8797936 in the aft end and the slotted spacer propellant, part no. 9328587 in the forward (delay assembly end) section of the tube, casing, part no. 8797921.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment Section 6 of MIL-A-48078 and 6.2 herein.

4.5 Test methods and procedures.

4.5.1 Moisture content.

4.5.1.1 Black powder. The moisture content of the black powder shall be determined in accordance with MIL-P-223.

4.5.1.2 Illuminant composition. The Karl Fischer Method as stated in MIL-STD-1234, Method 101.2 up to paragraph 5.3 shall be used. A fifty gram sample plus or minus 0.1 gram shall be added to a 500 milliliter (ml) volumetric flask containing approximately 300 to 400 ml of methanol and 25 grams of dry sodium nitrate. The flask shall be stoppered and the contents swirled cautiously for several minutes until the material is thoroughly dispersed. The sample shall be allowed to remain in contact with the methanol for approximately two hours. Then the 500 ml volumetric flask shall be filled up to the 500 ml mark with methanol and swirled again. A blank without the sample shall be put through the same procedure. After the sample has settled, a 50 ml aliquot of the clear supernatant liquid shall be withdrawn and immediately, transferred to the standard titration vessel containing approximately 100 ml of methanol which has just been titrated to the preliminary end point as described in MIL-STD-1234, Method 101.2, paragraph 5.1. The final end point shall be reached in 3.5 minutes in the manner described in MIL-STD-1234, Method 101.2 paragraph 5.4. A 50 ml aliquot of the blank shall be titrated in the same manner. The water content shall be calculated as follows:

Percent water equal (VR-S) - (V'R-S')
Multiply the result by 100F and divide by W

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Where:

F = grams of water per ml of standard water in methanol solution
V = ml of Karl Fischer reagent added to the sample
V' = ml of Karl Fischer reagent added to the blank
R = ml of standard water in methanol solution per ml fo Karl Fischer reagent
S = ml of standard water in methanol solution for titration of sample
S' = ml of standard water in methanol solution with back titration of blank
W = weight of sample in grams

4.5.1.3 Illuminant composition with VAAR (when applicable). The moisture and volatiles content shall be determined by placing an accurately weighed portion of approximately 25 grams (gr) of illuminant composition in a dried tared aluminum weighing dish, of sufficient diameter, to a depth of not more than 0.4 inch. A polytetrafluoroethylene cover shall be placed on the dish and the dish, contents and cover accurately weighed. The cover shall be removed and the dish and contents heated at 95 plus or minus 2 degrees Centigrade (C) for 2 hours. The dish shall be removed from the heat and then cooled to room temperature in a dessicator. After cooling, the cover shall be replaced and the dish, contents and cover shall be accurately weighed. The loss in weight shall be calculated as the percent of moisture and volatiles using the following method:

Percent moisture and volatiles equals 100 (A minus B) divided by C

Where:

A = weight of dish, cover and contents before drying
B = weight of dish, cover and contents after drying
C = weight of sample before drying

4.5.1.4 Delay composition. An accurately weighed portion of approximately 5 grams (gr) of delay composition shall be placed in dried tared aluminum weighing dish 2 inches in diameter. A polytetrafluorethylene cover shall be placed on the dish and the dish, contents and cover accurately weighed. The cover shall be removed and the dish and contents heated at 95 plus or minus 2 degrees Centigrade (C) for 2 hours min. At the end of the 2 hour drying period, remove the dish from the heat, replace the cover, and cool to room temperture in a desiccator. After cooling, accurately weigh the dish, cover and contents. The loss in weight shall be calculated as the percent of moisture using the following method:

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Percent moisture equals 100 (A minus B) divided by C.

Where:

A = weight of dish, cover and contents before drying
B = weight of dish, cover and contents after drying
C = weight of sample before drying

4.5.1.5 Felt, fiberboard, cork, wood, paper products and parachute cloth. An accurately weighed portion of at least 10 grams of the material weighed to the nearest milligram shall be transferred to a tared weighing dish and placed in an oven and heated at a temperature of 100 degrees plus or minus 2 degrees Centigrade (C) for 2 hours. The dish and contents shall be removed from the oven and placed in a desiccator, cooled, and weighed. The difference in weight shall be calculated to percentage of moisture in the sample.

4.5.1.6 Propellant assembly, propellant grain. The moisture content of the propellant assembly and propellant grain shall be determined in accordance with the procedure specified in MIL-P-223, or an approved alternate.

4.5.2 Functioning. The following functioning tests, in accordance with Acceptance Test Procedure S1-GH-VAR Supplement S1-GH- (latest revision), shall be performed at a Government Proving Ground.

4.5.2.1 Hot temperature - (68° to 74°C or 155 to 165°F). The packed signals shall be conditioned for a minimum of 16 hours at the above temperature. The packed signal shall be taken from the conditioning chamber, removed from the metal container and fired vertically from a fixture. (See 4.4.4) The performance of the sample shall be observed for compliance with the requiements.

NOTE: Assembly must be fired within 60 seconds of removal from the conditioning chamber.

4.5.2.2 Ambient - (18° to 24°C or 65° to 75° F). The signal packed in sealed container shall be conditioned for a minimum of 16 hours at the above temperature. After conditioning, the sample shall be removed from the container and fired from a fixture (See 4.4.4). The functioning performance shall be observed for compliance with the requirements.

NOTE: Samples must be fired within 60 seconds of removal from the conditioning chamber.

4.5.2.3 Cold temperature - (-57°C to -51°C or -70° to -60°F).

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4.5.2.3.1 Fixture fired. The packed signals shall be conditioned for a mimimum of 16 hours at cold temperature. The packed signal shall be taken from the conditioning chamber, removed from the metal container and fired vertically from a fixture (See 4.4.4). The functioning performance of the sample shall be observed for compliance with the requirements.

NOTE: Samples must be fired within 60 seconds of removal from the conditioning chamber.

4.5.2.3.2 Hand fired. The packed signals shall be conditioned for a minimum of 16 hours at cold temperature. The packed signal shall be taken from the conditioning chamber, removed from the metal container and fired vertically by hand (See 6.4). The performance of the signal shall be observed for compliance with the requirements.

NOTE: Samples must be fired within 60 seconds of removal from the conditioning chamber.

4.5.2.4 Test validity. If for any reason the proving ground considers that the test conditions have detrimentally affected the test results, the test shall be declared invalid and a new test performed with additional samples.

4.5.3 Leakage. The packed sealed container shall be placed in the test cylinder. Apply air in the cylinder to a pressure of 3 to 5 pounds per square inch (p.s.i.) and hold for eight (8) seconds min. Observe pressure indicatior for a drop in pressure during the holding time. Any assembly that fails to comply with the requirement shall be classed defective and removed from the lot.

4.5.4 Container tear strip test. The tear strip of the container shall be tested in accordance with the procedures specified in MIL-C-10464 for compliance with the requirements specified. If any container fails to comply with the requirements, it shall be classed defective.

4.5.5 Density of propellant grain.

4.5.5.1 Overall density. The propellant grain shall be accurately measured, the volume determined and converted to cubic centimeters. The grain shall then be accurately weighed and the density in grams per cubic centimeter calculated and compared with the requirements. Any propellant grain that fails to comply with the requirements shall be classed defective and removed from the lot.

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4.5.5.2 Density of any section. Each of the grains shall be cut into two (2) approximately equal size cylinders. Each cylinder shall be accurately weighed. Fill the beaker with clean mercury, insert the immersing fixture and level off the mercury by pressing down with the flat glass plate until it is even with the top of the beaker. Then place the cylinder beneath the prongs of the immersing fixture and the level the mercury by pressing down with the flat glass until it is even with the top of the beaker. Catch the excess mercury in a clean evaporating dish. Weigh the mercury displaced by the cylinder. Calculate the density of the cylinder as follows:

Density = 13.54 W divided by the weight of the displaced mercury grams.

Where:

W = weight of the cylinder in grams

If any section of the grain fails to comply with the requirements, the lot shall be rejected.

4.5.6 Hydrostatic test of tube, casing. The casing tube shall be tested in a hydrostatic test device (See 4.4.4). The minimum pressure shall be applied for not less than 5 seconds. The tube shall be observed for evidence of failure to comply with the requirements. Any casing tube that bursts, cracks or shows evidence of distortion shall be classed defective and removed for the lot.

4.5.7 Bolt pull strength. The bolts shall be tested in accordance with the procedure and equipment specified in 4.4.4. If any bolt fails to comply with the applicable drawing requirement, it shall be classed defective.

4.5.8 Hardness of tail vane. The tail vane shall be tested for hardness as specified in Federal Standard No. 151, Method 243. If any tail vane fails to comply with the requirement, the heat treated batch represented by the sample shall be rejected.

4.5.9 Push test of firing cap and spring assembly. The firing cap and spring clip shall be tested in accordance with the equipment and procedure specified in 4.4.4. If any assembly fails to comply with the applicable requirements, it shall be classed defective.

4.5.10 Candlepower and color values. The illuminant assemblies shall be tested for candlepowder and color value in accordance with the procedures and equipment specified in dwg. 9201136, 9201268, 9201390 and associated dwgs., 9201392 and 9247071. Any sample failing to comply with the applicable requirement shall be classed defective.

NOTE: Illuminant assemblies from each mix, batch or sublot making up the inspection lot shall be uniformly represented in the above sample. If the inspection lot fails this test, each sublot indicating failure shall be rejected. Each of the remaining sublots shall be treated as an individual respective inspection lot and be re-tested according to the above procedures.

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4.5.11 Weld peel test. The weld on the tail assembly shall be tested in accordance with the procedures specified in MIL-W-12332. After test the assembly shall be examined to determine compliance with the drawing requirements. If any weld fails to comply with the requirements, the lot shall be rejected.

4.5.12 Propellant ballistic test. The signals shall use propellant spacer, part no. 8797936, in the forward and aft end while following the test procedure of 4.5.2. Simulated or dummy payloads may be used.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A. The signals shall be preserved and packaged in accordance with Drawing 7548414.

5.2 Packing.

5.2.1 Level A. The containers shall be packed and packaged in accordance with Drawing 7548415.

5.3 Marking. The containers shall be marked as specified on Drawings 7548414 and 7548415.

6. NOTES

6.1 Ordering data. Procurement documents shall specifiy the following:

a. Title, number and date of this specification.
b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified in MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Submission of designs for approval. Contractor designs shall be approved by the Government prior to fabrication or procuring the equipment. Designs shall be submitted for approval to ARRADCOM, DRDAR-QAR-I, Dover, New Jersey 07801 in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. Partial submission of inspection equipment designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. ARRADCOM design review will normally be accomplished within one month after receipt.

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6.3 Alternate test methods. Prior approval of the Contracting Officer is required for use of equivalent test mehtods. A description of the proposed method should be submitted thru the Contracting Officer to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-QAR-I, Dover, New Jersey 07801. This description should include but not be limited to the procedures used, the accuracy and precision of the method, test data to demonstrate the accuracy and precision and drawings of any special equipment required.

6.4 Hand firing. Personnel hand firing signals after cold temperature conditioning should wear protective gloves and other suitable protective clothing.

6.5 Drawings. Drawings listed in Section 2 of this specification under the heading, "US ARMY ARMAMENT RESERACH AND DEVELOPMENT COMMAND (ARRADCOM)" may also include drawings prepared by and identified as Edgewood Aresenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technial data originally prepared by these activities is now under cognizance of ARRADCOM.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with repsect to the previous issue, due to the extensiveness of the changes.

6.7 Proving ground functioning test summary.

Firing Method & Conditioning Temp	First Article	Production Lots: First Three Consecutive	Production Lots: Remaining Acceptable	Requirement
Fixture				
68 to 74°C (155 to 165°F)	32	100	25	See 3.4.3
18 to 24°C (65 to 75°F)	32	125	50	See 3.4.4
-57 to -51°C (-70 to -60°F)	16	32	10	See 3.4.5
Hand:				
-57 to -51°C (-70 to -60°F)	16	125	50	See 3.4.5

Custodian:
Army-AR

Preparing Activity:
Army-AR

Project No: 1370-A-067

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MIL-S-13257H Signals, Illumination, Ground Red Star, M126A1

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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